Retail Sales Performance Analysis

Techbridge Services, Task 8

Prepared by Adetola Temitope Samuel | temitopeadetola48@gmail.com

Project Objective

This project looks at how the business performed in sales and customer activity from January 2026 to June 2026. It answers these questions:

1. How much revenue did the business make?
2. How many product units were sold?
3. How many customers bought from the business (20 customers are on the customer list)?
4. Which customer brought in the most revenue?
5. How did each city perform?
6. Which type of customer bought the most?

Dataset

The dataset has three sheets: Customers, Products and Transactions.

- **Customers** lists the 20 customers who made purchases, with their ID, name, city and customer type.
- **Products** lists the 10 products on sale, with their ID, name, category and unit price.
- **Transactions** lists the 150 sales, with the transaction ID, customer ID, product ID, date, quantity sold and discount given.

Product ID connects the Transactions sheet to the Products sheet, and Customer ID connects the Transactions sheet to the Customers sheet. Because of these links, I merged all three sheets into one sheet, so every sale shows its price, product, customer, city and customer type in one place. This makes the analysis clearer and more accurate.

Data Cleaning

1. **Merging the sheets.** I used the VLOOKUP function to bring the product and customer details into the Transactions sheet.

2. **Missing customer.** One sale (T0040) uses customer ID C999, which is not on the customer list, so its customer name, city and customer type came back empty. I used IFERROR to label these cells NOT FOUND. This sale still has a price, so its revenue of 450,000 is kept in the totals.
3. **Missing product.** One sale (T0080, customer Aisha Bello) uses product ID P999, which is not on the product list, so the product, category and unit price were missing. Its revenue could not be worked out, so I left it out of all totals.
4. **Extra spaces.** I found a double space in one column and used the TRIM function to clean it.
5. **Negative quantity.** One order (T0120) was recorded as -2. A quantity cannot be negative, so I treated it as a data entry mistake and corrected it to 2. If it was a real return, total revenue would be 41,563,200.
6. **Revenue.** I worked out revenue by multiplying the unit price by the quantity.
7. **Discount.** The business gave discounts to some customers, so I added a Total Revenue column. It takes the discount off the revenue: Total Revenue = Quantity x Unit Price x (1 - Discount). Discounts are stored as fractions, so 0.05 means 5%.
8. **Month.** To study monthly sales, I added a Month column using the TEXT function.

After cleaning, 149 sales had revenue that could be calculated, covering 450 items sold.

Analysis

What this analysis is about

This analysis looks at the business's sales for the first six months of 2026. It shows how much money came in, what sold best, where it sold, and how sales changed from month to month.

Main numbers

1. The business made **43,363,200** in sales (revenue after discounts).
2. There were **149** sales.
3. **450** items were sold.
4. The average sale was worth **291,028**.
5. Discounts cost **751,800**, which is about 1.7% of what sales would have been without them.

What was found

1. **Best-selling products.** Monitor was the best product with 7,992,000 (18.4% of all sales). Printer (7,104,000), Laptop (6,570,000) and Tablet (6,512,000) came next. Together these four bring in about 65% of all sales, and with Desk (6,174,000) the top five bring in about 79%.
2. **Weakest products.** Mouse, Keyboard and Office Chair brought in the least. Mouse is the lowest with 235,200 (0.5%).
3. **Best category.** Electronics earned the most, 25,484,000 (58.8% of all sales). It is about three times bigger than the next category, Furniture (7,580,000, 17.5%). Office Equipment earned 7,104,000 (16.4%).
4. **Weakest category.** Accessories earned the least, 3,195,200 (7.4%). Many accessories were sold (about 40% of all items), but they are cheap, so they add little money.
5. **Best cities.** Kano earned the most with 7,222,000 (16.7%), followed by Abuja with 7,071,000 (16.3%) and Lagos with 6,473,250 (14.9%). Together these three bring in about 48% of all sales. Lagos has the most sales (21), but its average sale (308,250) is lower than Kano's (451,375). Kaduna has only 7 sales, but each one is large (average 476,571).
6. **Weakest cities.** Enugu performed the least with 125,600 (0.3%), along with Abeokuta and Calabar.
7. **Customer types.** Retail customers brought in 23,975,850 (55.3%), Corporate customers 16,320,350 (37.6%) and Wholesale customers 2,617,000 (6.0%). Wholesale customers buy often but spend only about 90,241 each time.
8. **Best customers.** The top five customers bring in 19,000,000 (43.8% of all sales). Ibrahim Sani (Abuja) is first with 4,716,000, followed by Hauwa Ali (3,768,000), Tunde Adebayo (3,726,000), Mariam Yusuf (3,454,000) and Fatima Musa (3,336,000). Kelvin George is the lowest with 109,600.
9. **Sales by month.** January was the best month with 9,642,250 (22.2%), followed by May with 9,167,500 (21.1%) and March with 8,660,250 (20.0%). February (5,319,850), April (5,289,950) and June (5,283,400) were much lower, at about 12% each.
10. **Number of sales.** The number of sales stayed about the same every month (around 25). So the ups and downs come from the size of the orders, not the number of orders.

Key Insights

1. **Electronics makes the most money.** Out of every 100 naira earned, about 59 came from Electronics. Monitors, Printers, Laptops and Tablets are the best sellers.
2. **Three cities bring in nearly half of the sales.** Kano, Abuja and Lagos are the best places. Cities like Enugu, Abeokuta and Calabar bring in very little.

3. **Sales are good one month and weak the next.** January, March and May did well. February, April and June were weaker. The number of sales was about the same each month, so the difference is that some months had bigger orders.
4. **Cheap items sell a lot but earn little.** Accessories, like mouse and keyboard, are 4 out of every 10 items sold, but they bring in less than 1 out of every 10 naira. Selling them together with bigger items would help.
5. **A few customers matter a lot.** The top five customers bring in about 44% of all sales.

SQL Analysis

What I did

I used SQL (in PostgreSQL) to answer simple questions: how much money did the business make, from which products, in which places, and from which customers? First I created three tables (Customers, Products and Transactions). Then I loaded the data into them and wrote queries to study the sales.

JOINs

The sales information was spread across different tables, so I had to connect them. This is called a JOIN.

1. The Transactions table shows what was bought and how many, but not the price. So I joined it to the Products table using the product ID to get the price.
2. The Transactions table also does not show the customer's city or type. So I joined it to the Customers table using the customer ID.

I used a normal join (an inner join), which keeps only the rows that match in both tables. Because of this, the two sales with unknown codes (one customer code and one product code that are not on the lists) are left out of some results.

GROUP BY

GROUP BY puts rows into groups so they can be added up group by group. I grouped the sales by product, product category, customer, city and customer type (Retail, Corporate and Wholesale).

Aggregations

Aggregations are calculations on many rows at once. I used two:

1. **SUM** to add up total revenue. For each sale, revenue is quantity x price x (1 - discount). I then added these up for each group.

2. **AVG** to find the average revenue per sale.

I also used ORDER BY to arrange the results from highest to lowest.

Question 1: What is the total revenue generated by each product?

```
select p.product_name, p.product_id,
sum(t.quantity * p.unit_price * (1 - t.discount)) as total_revenue
from transactions as t
join products as p on t.product_id = p.product_id
group by p.product_name, p.product_id
order by total_revenue desc;
```

Result: Monitor leads with 7,992,000, followed by Printer with 7,104,000. Mouse generated the least with 235,200.

Question 2: Which product category generated the highest revenue?

```
select p.category,
sum(t.quantity * p.unit_price * (1 - t.discount)) as total_revenue
from transactions as t
join products as p on t.product_id = p.product_id
group by p.category
order by total_revenue desc;
```

Result: Electronics earned 25,484,000. Accessories earned the least, 3,195,200.

Question 3: Which customers generated the most revenue?

```
select c.customer_name, c.customer_id,
sum(t.quantity * p.unit_price * (1 - t.discount)) as total_revenue
from transactions as t
join customers as c on t.customer_id = c.customer_id
join products as p on t.product_id = p.product_id
group by c.customer_name, c.customer_id
order by total_revenue desc;
```

Result: Ibrahim Sani contributed the most with 4,716,000, then Hauwa Ali with 3,768,000 and Tunde Adebayo with 3,726,000. Kelvin George had the least with 109,600.

Question 4: Which city generated the highest revenue?

```
select c.city,
sum(t.quantity * p.unit_price * (1 - t.discount)) as total_revenue
from transactions as t
join customers as c on t.customer_id = c.customer_id
join products as p on t.product_id = p.product_id
group by c.city
order by total_revenue desc;
```

Result: Kano generated 7,222,000. Enugu generated the least, 125,600.

Question 5: What is the average transaction value?

```
select avg(t.quantity * p.unit_price * (1 - t.discount)) as
avg_transaction_value
from transactions as t
join products as p on t.product_id = p.product_id;
```

Result: The average sale is about 291,028.

Important SQL findings

1. Total revenue is 43,363,200 from 149 sales, and the average sale is 291,028.
2. Best products: Monitor (7,992,000), Printer (7,104,000), Laptop (6,570,000) and Tablet (6,512,000). Mouse earned the least (235,200).
3. Best category: Electronics earned 25,484,000, about 58.8% of all sales. Accessories earned the least (3,195,200).
4. Best cities: Kano (7,222,000), Abuja (7,071,000) and Lagos (6,473,250). Enugu had the least (125,600).
5. Best customer: Ibrahim Sani from Abuja, with 4,716,000.
6. Customer types: Retail brought in the most (23,975,850), then Corporate (16,320,350). Wholesale brought in the least (2,617,000).

Dashboard

I built a Power BI dashboard to show the results at a glance. It has four key numbers at the top: Total Revenue (43,363,200), Total Transactions (149), Total Units Sold (450) and Average Transaction Value (291,028). Below them are four charts: Top Products by Revenue, Revenue by Product Category, Revenue by City and Monthly Revenue Trend.

Recommendations

1. Keep Monitors, Printers, Laptops and Tablets well stocked and promote them most, because Electronics earns the most money.
2. Sell accessories together with big items, such as a mouse with a laptop, to raise the value of each order.
3. Run special offers in February, April and June to lift the slow months.
4. Look after Kano, Abuja and Lagos, and get Lagos customers to buy bigger items. Grow sales in cities like Kaduna, where each sale is large.
5. Reward the top customers to keep them, and keep finding new customers.
6. Review wholesale prices and the products wholesale customers choose, since they spend little each time.
7. Check that every sale has a valid customer ID and product ID before it is saved.

Data Notes

1. Transaction T0080 (product ID P999) was left out of all totals because the product is not on the product list.
2. Transaction T0040 (customer ID C999) is included in revenue (450,000) but has no city or customer type. It is left out of the city, customer and customer type results, which therefore add up to about 450,000 less than total revenue.
3. Transaction T0120 was recorded as -2 and treated as 2, as a data entry mistake. If it was a real return, total revenue would be 41,563,200.

Conclusion

1. Electronics and four products (Monitor, Printer, Laptop and Tablet) drive about 65% of revenue.
2. Three cities (Kano, Abuja and Lagos) generate almost half of all sales.
3. Revenue swings between strong and weak months because of large orders.